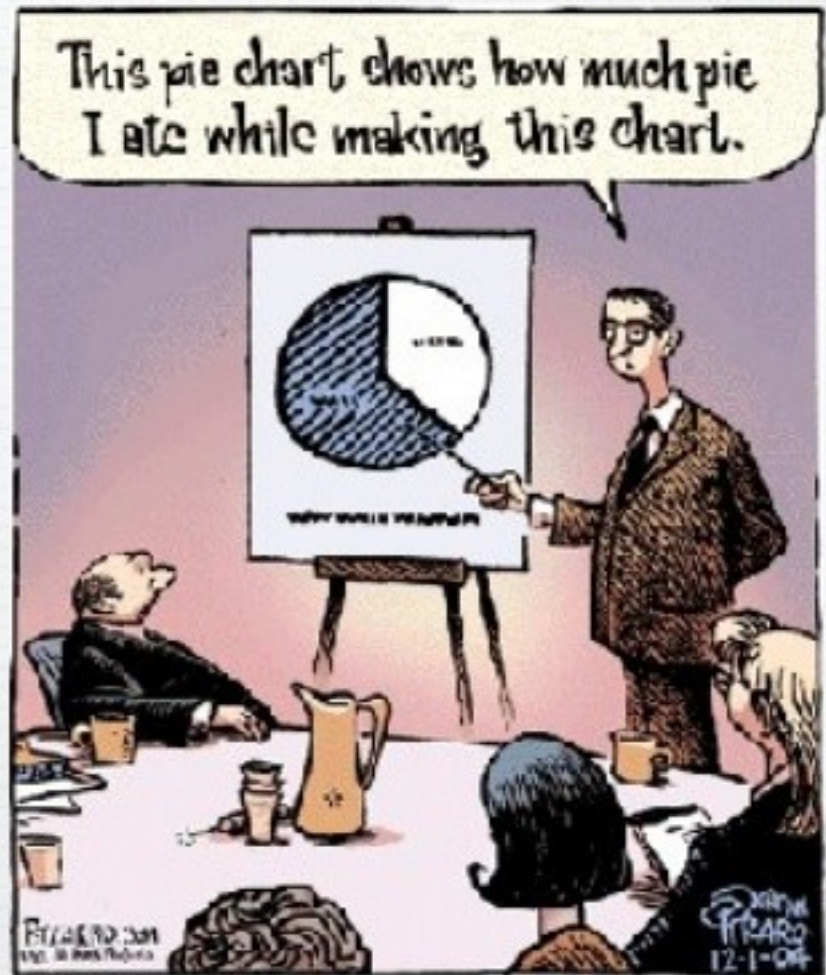


Data Analysis

Understanding your results by making
a graph to analyze your data

Every
graph tells
a story

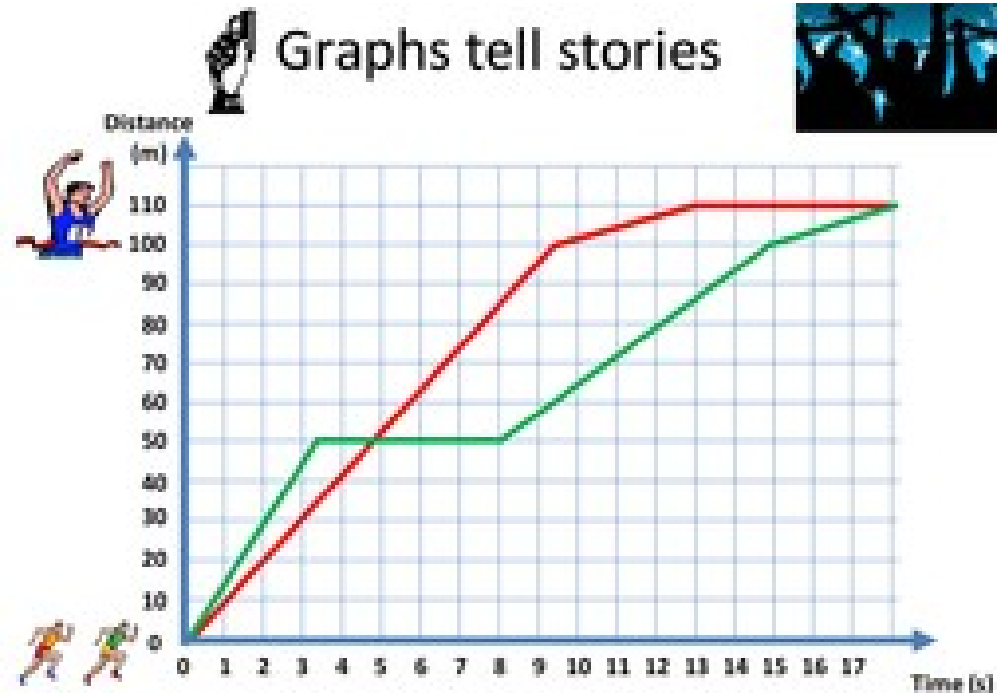
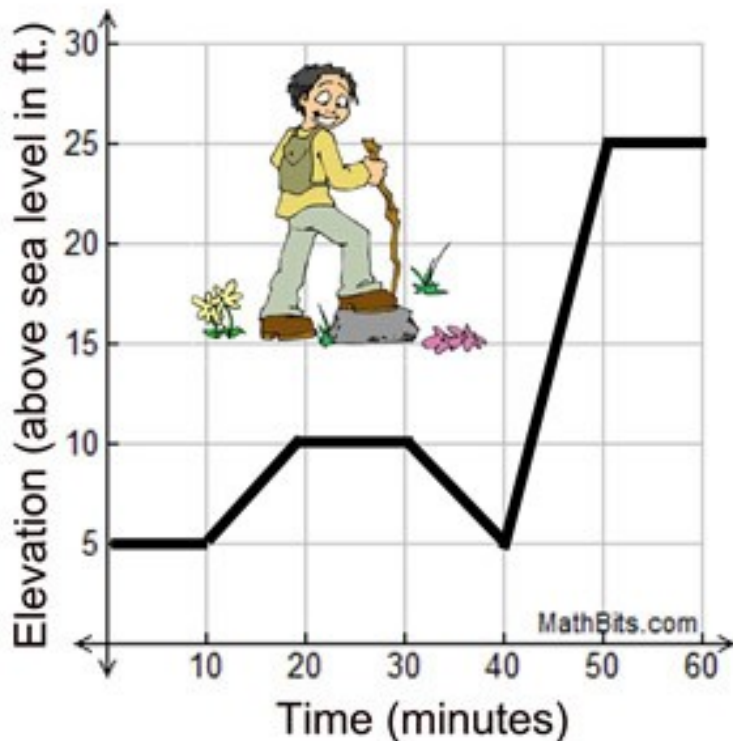


Goals for today

- Learn how to make a Graph from data you have collected
 - Data tables, labels, type
- Learn how to describe a graph
 - linear, non-linear, exponential
- Learn how to interpret a graph
 - What is the relationship? – proportional, directly proportional, inversely proportional, negative relationship, positive relationship
 - What physical quantity does the slope tell you?

Why Use Graphs

- Organizing our data into a graph does more than make it aesthetically pleasing (pretty)
- It tells a story



Data Tables

- Make a data table and then use it to plot your data

X	Y
0	0
1	2
2	4
3	6

IV	DV
0	0
1	2
2	4
3	6

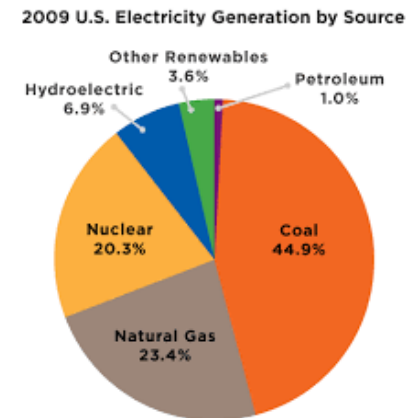
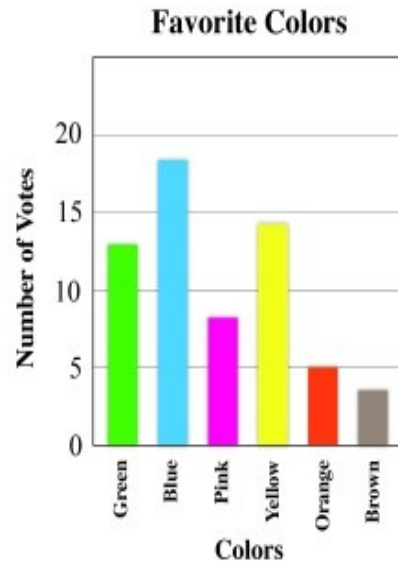
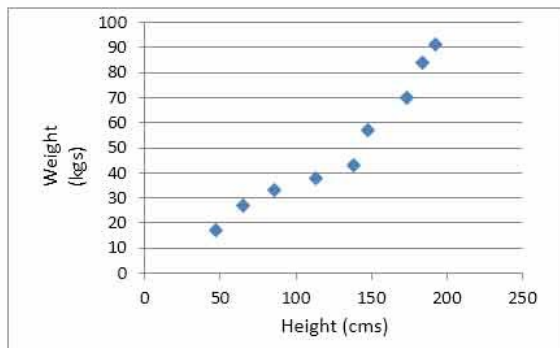
Time (second s)	Distanc e (meters)
0	0
1	2
2	4
3	6

t (s)	d (m)
0	0
1	2
2	4
3	6



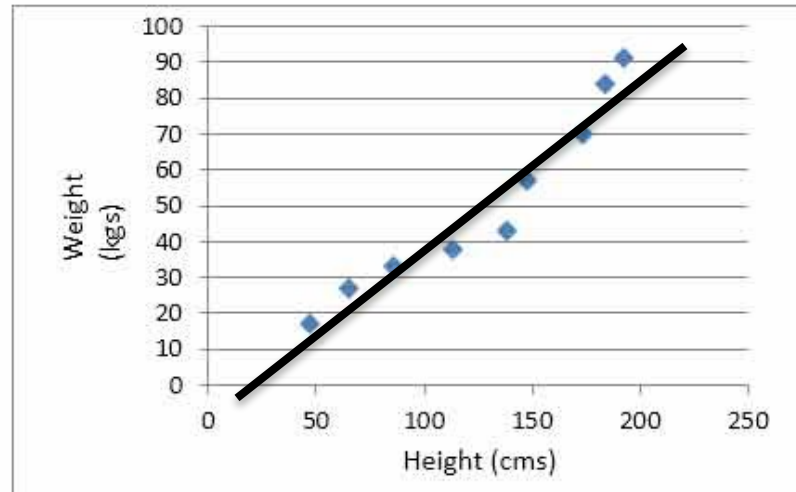
Graph Types

- X-Y Scatter Plot
- Bar/Column Graph
- Pie Chart



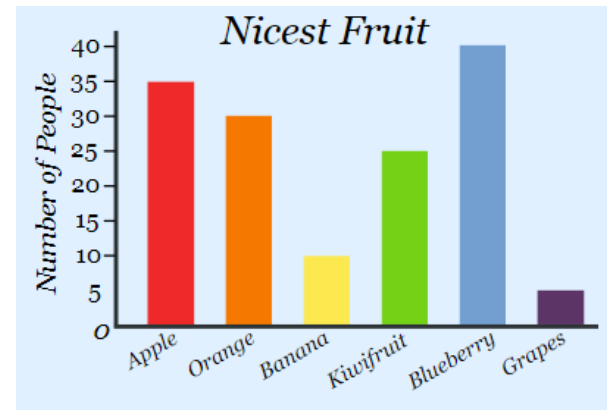
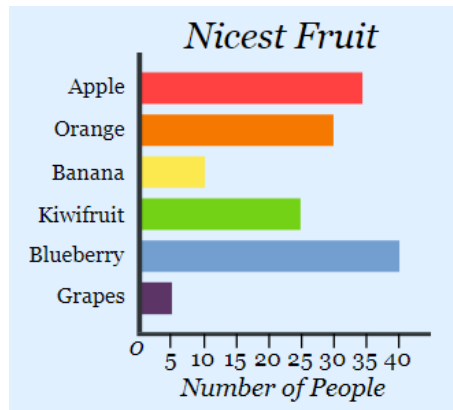
X-Y Scatter Plot

- used to determine **relationships** between the two different variables
- X-axis – Independent Variable
- Y- axis – Dependent variable
- most common type of graph you will use in science
- Do not connect the dots, instead use a trendline!



Bar/Column Graph

- used to **compare** two or more different groups
- they do not show changes over time
- There is no relationship

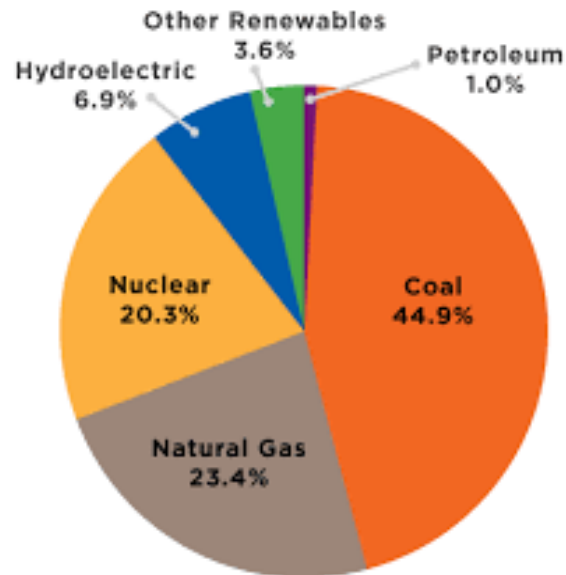


Both the **Bar** and the **Column** charts display data using rectangular **bars** where the length of the **bar** is proportional to the data value. Both are used to compare two or more values. However, their **difference** lies in their orientation. A **Bar chart** is oriented horizontally whereas the **Column chart** is oriented vertically.

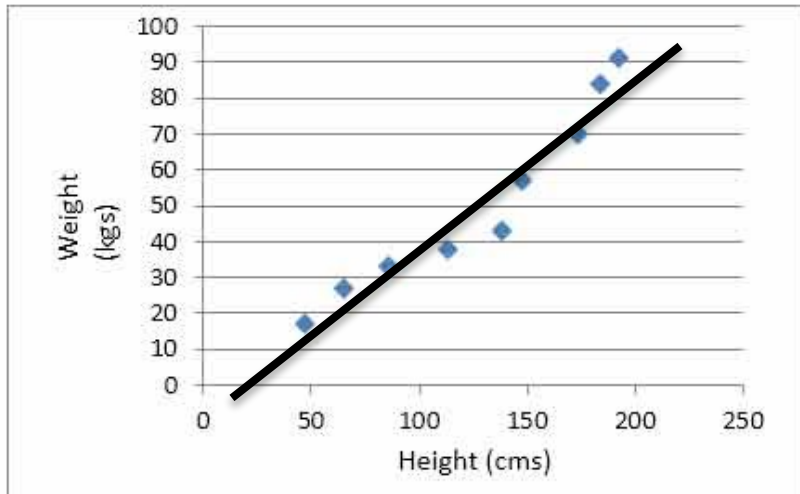
Pie Chart

- used when you are trying to compare parts of a whole
- they do not show changes over time
- There is no relationship

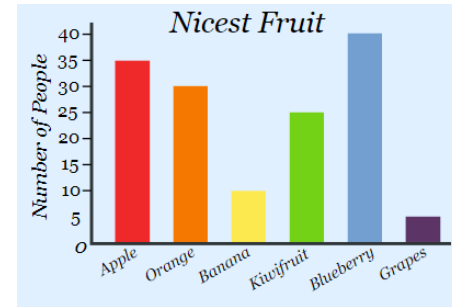
2009 U.S. Electricity Generation by Source



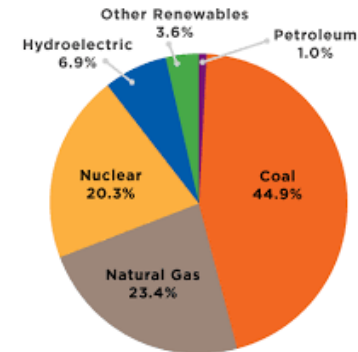
Where is the Relationship Shown?



RELATIONSHIP BETWEEN Y-
VALUE AND X-VALUE



2009 U.S. Electricity Generation by Source



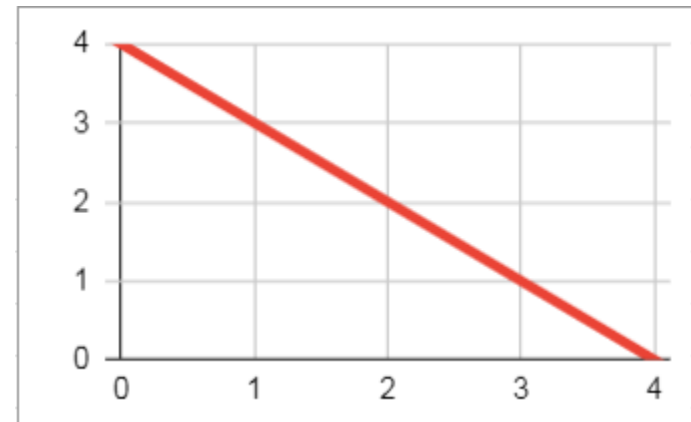
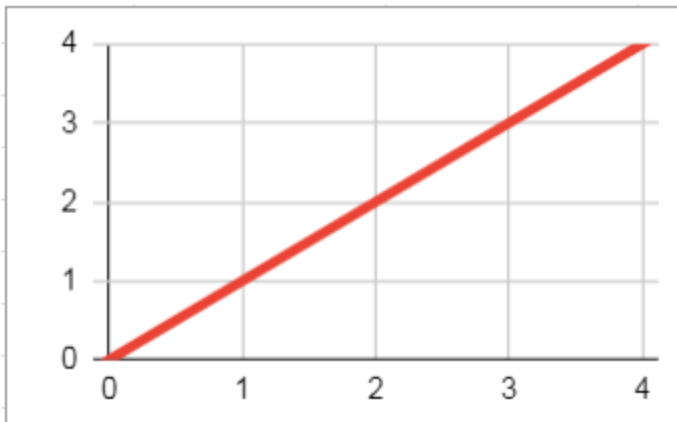
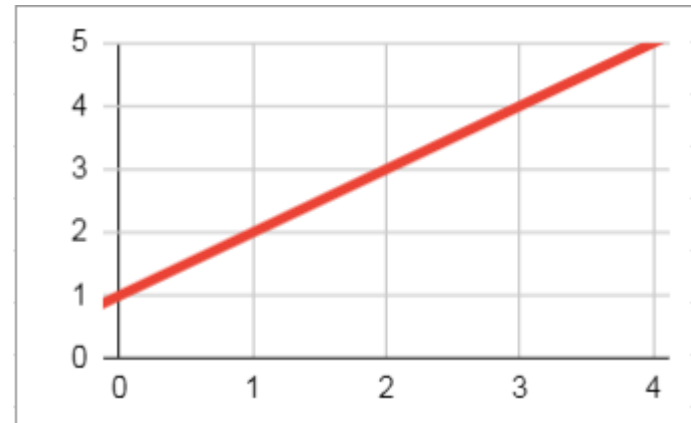
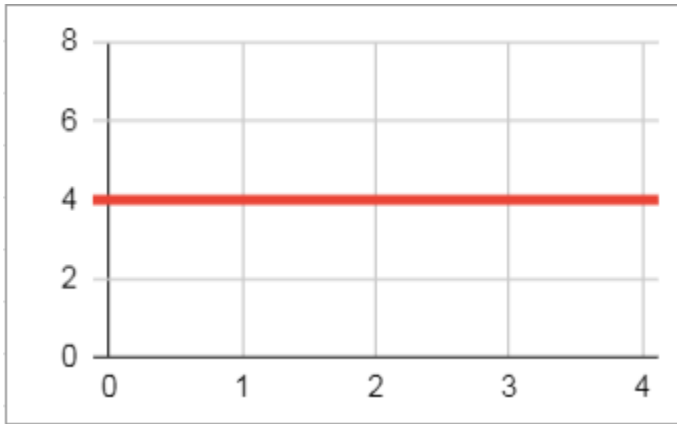
NO RELATIONSHIP

How to Interpret an X-Y Scatter Plot

- Is it:
 - Linear or non-linear
 - Proportional ,directly proportional or inversely proportional
 - Positive or negative relationship

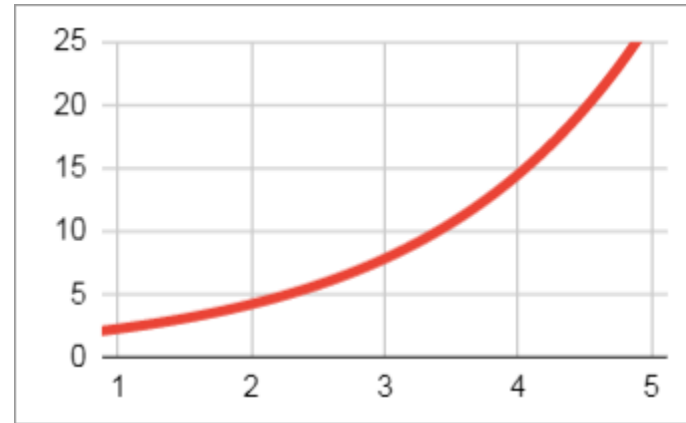
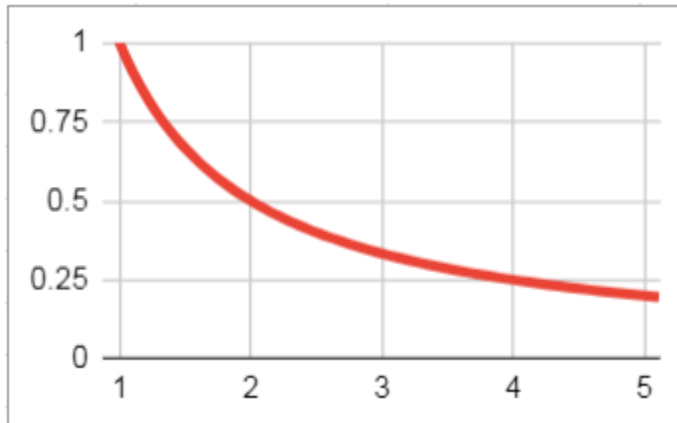
Linear & Constant Slope

- These types of graphs will always have straight lines



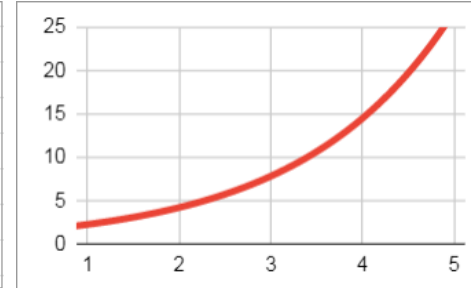
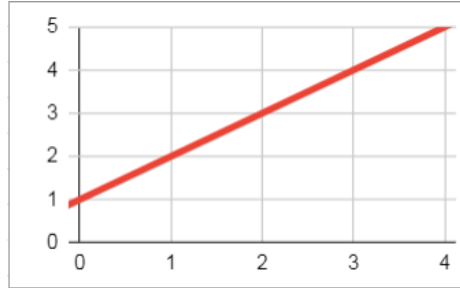
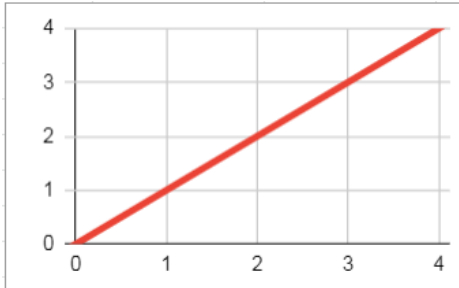
Non Linear & Changing Slope

- These types of graphs will always have curved lines

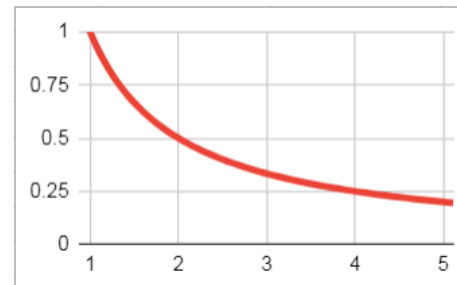
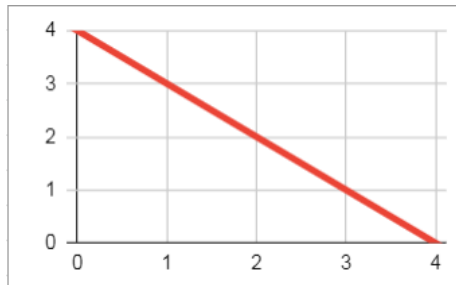


Positive or Negative Relationship

- Positive - as x-axis increases, y-axis increases too

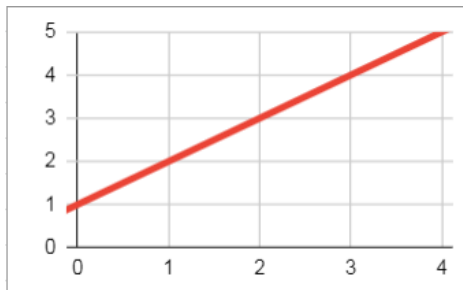


- Negative - as x-axis increases, y-axis decreases

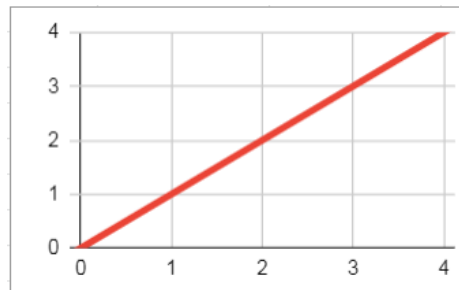


Proportionality

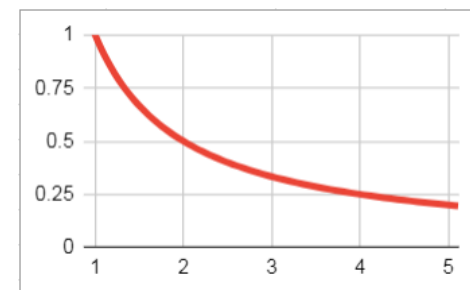
- Proportional – as IV increases, DV increases
- Directly Proportional – as IV increases, DV increases and goes through 0,0
- Inversely Proportional – as IV increases, DV decreases



PROPORTIONAL



DIRECTLY
PROPORTIONAL

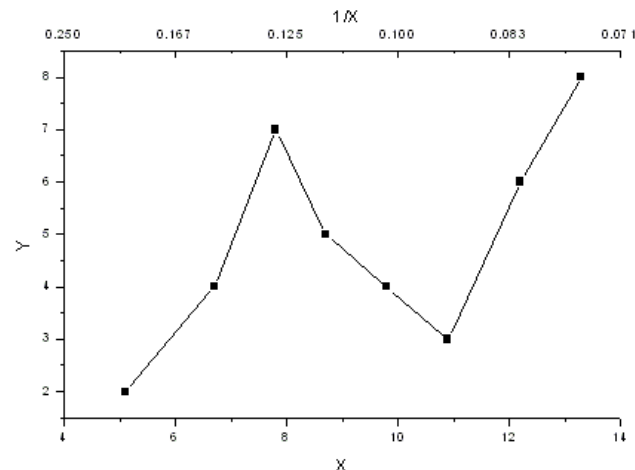
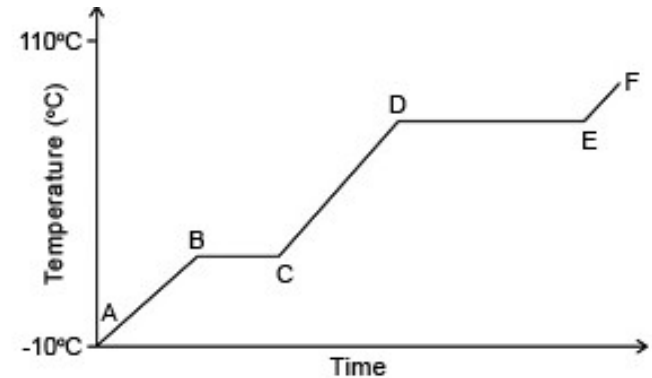
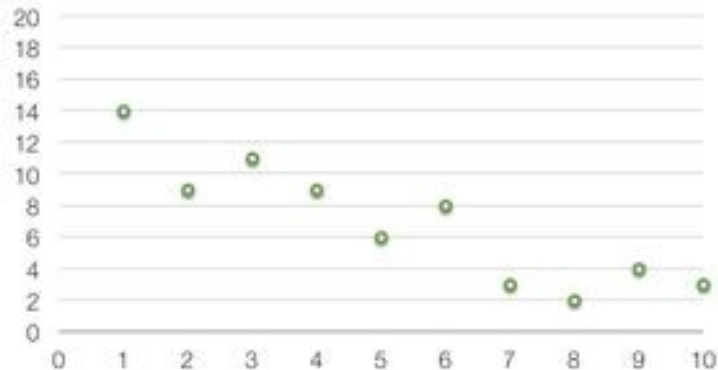


INVERSELY
PROPORTIONAL

Nonlinear

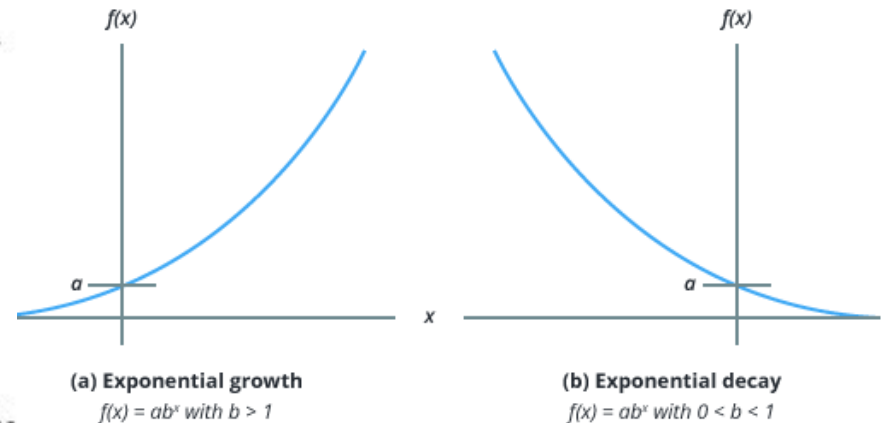
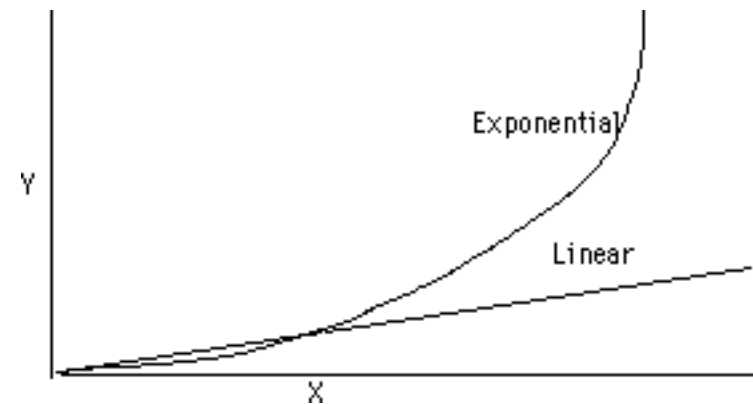
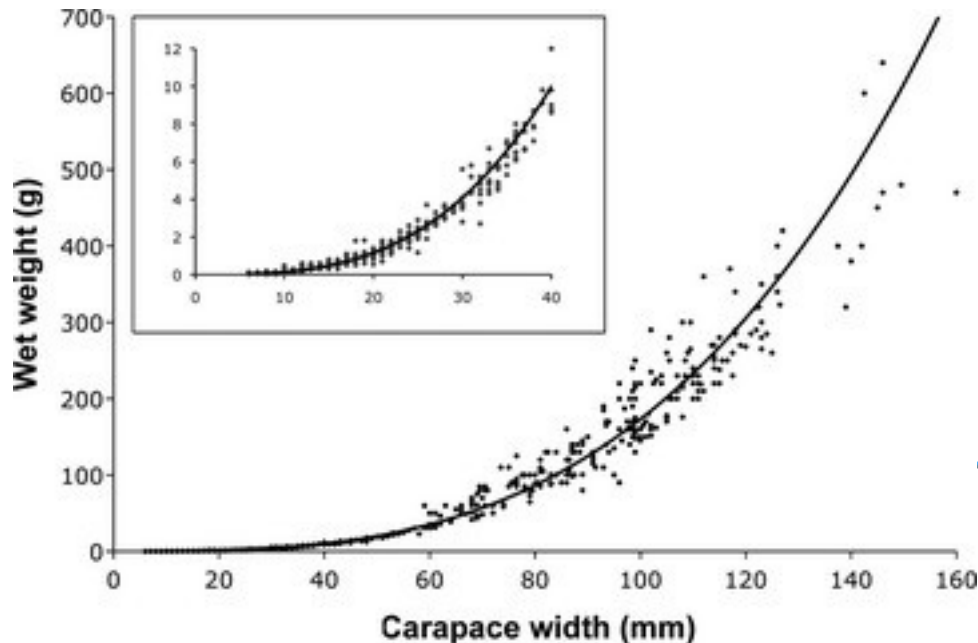
- Not a straight line path
- Look at the WHOLE picture for trends or patterns
 - may or may not be present

Example 3. Non-Linear Graph



Exponential

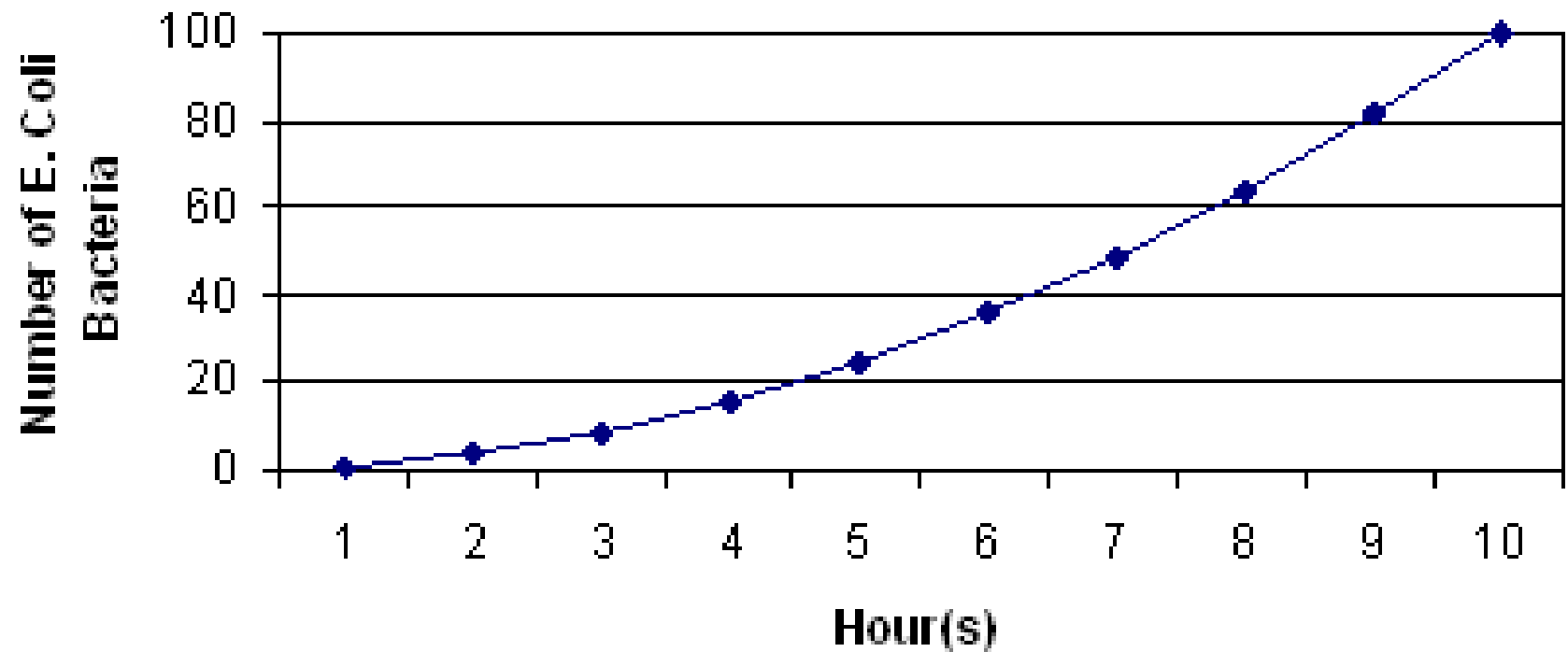
- Graph increases or decreases by huge factors (shaped as a curve)



Examples

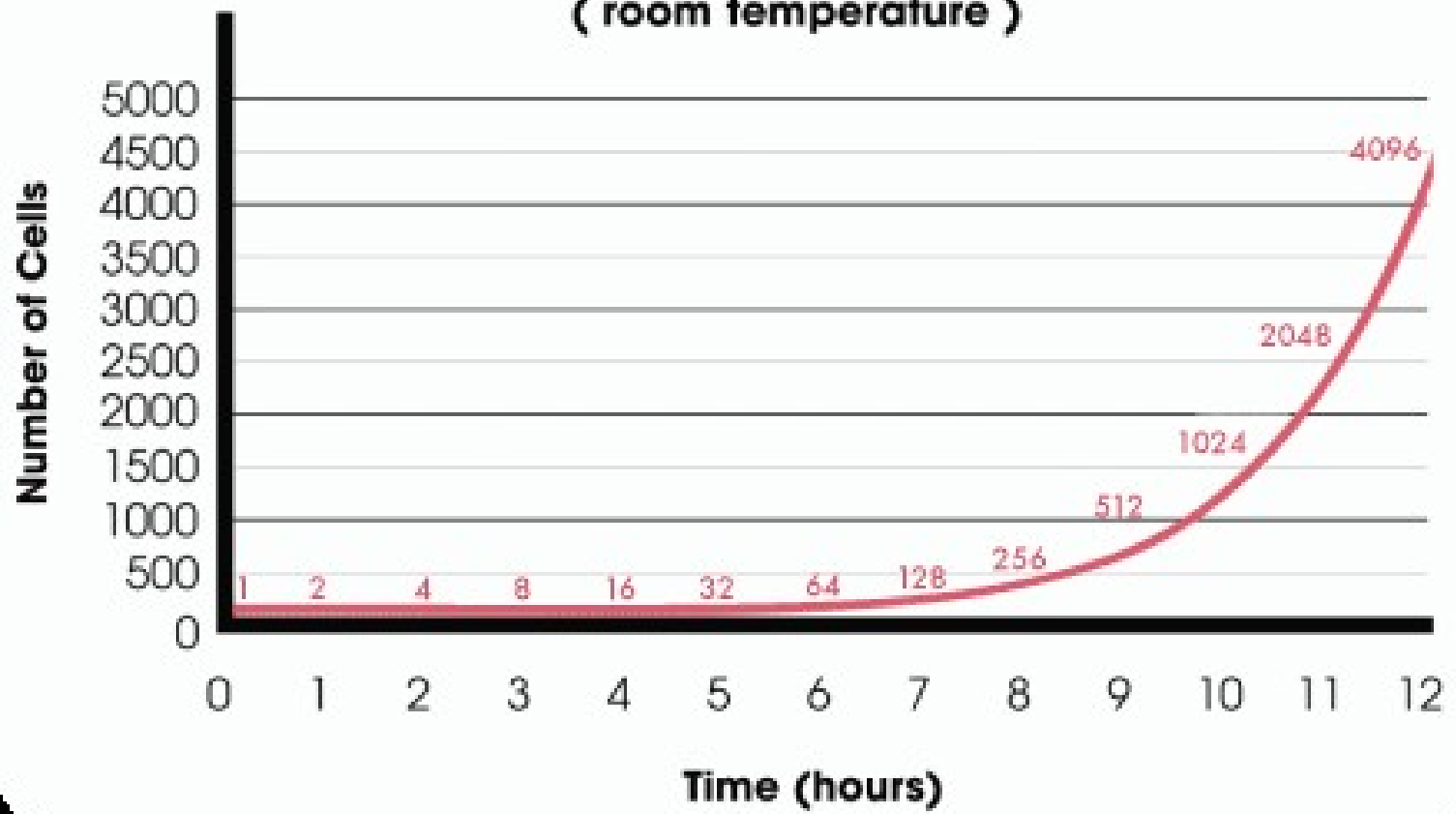
- Discuss the following graphs
 - What shape is the graph? (linear, nonlinear, exponential)
 - What is the relationship between variables? (positive or negative)
 - Is it proportional, directly proportional, inversely proportional?
 - What story does it tell? (as X increases, Y _____)

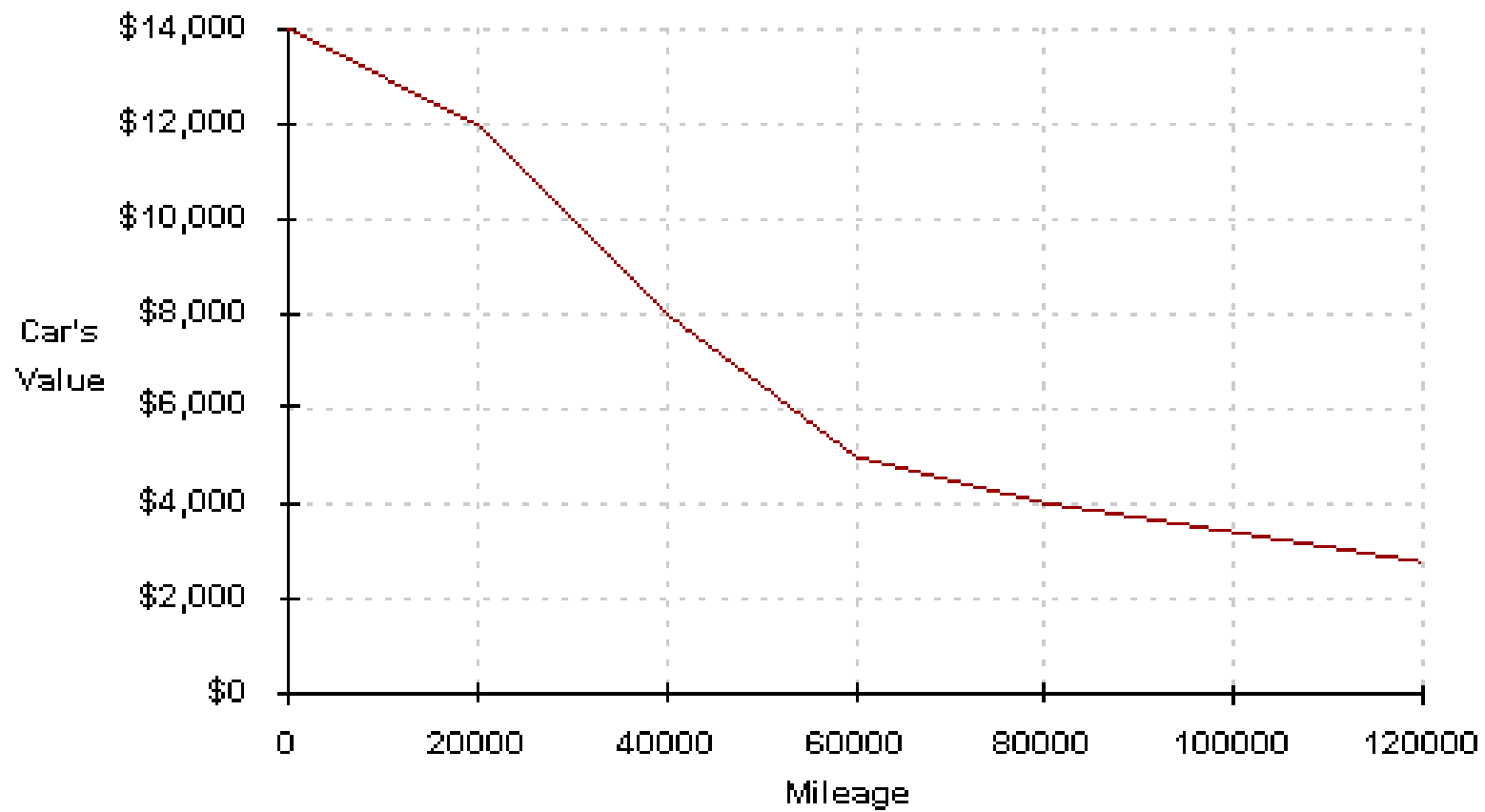
The Number of Reproducing E. Coli Bacteria Over Ten Hours in Agar at 25°C

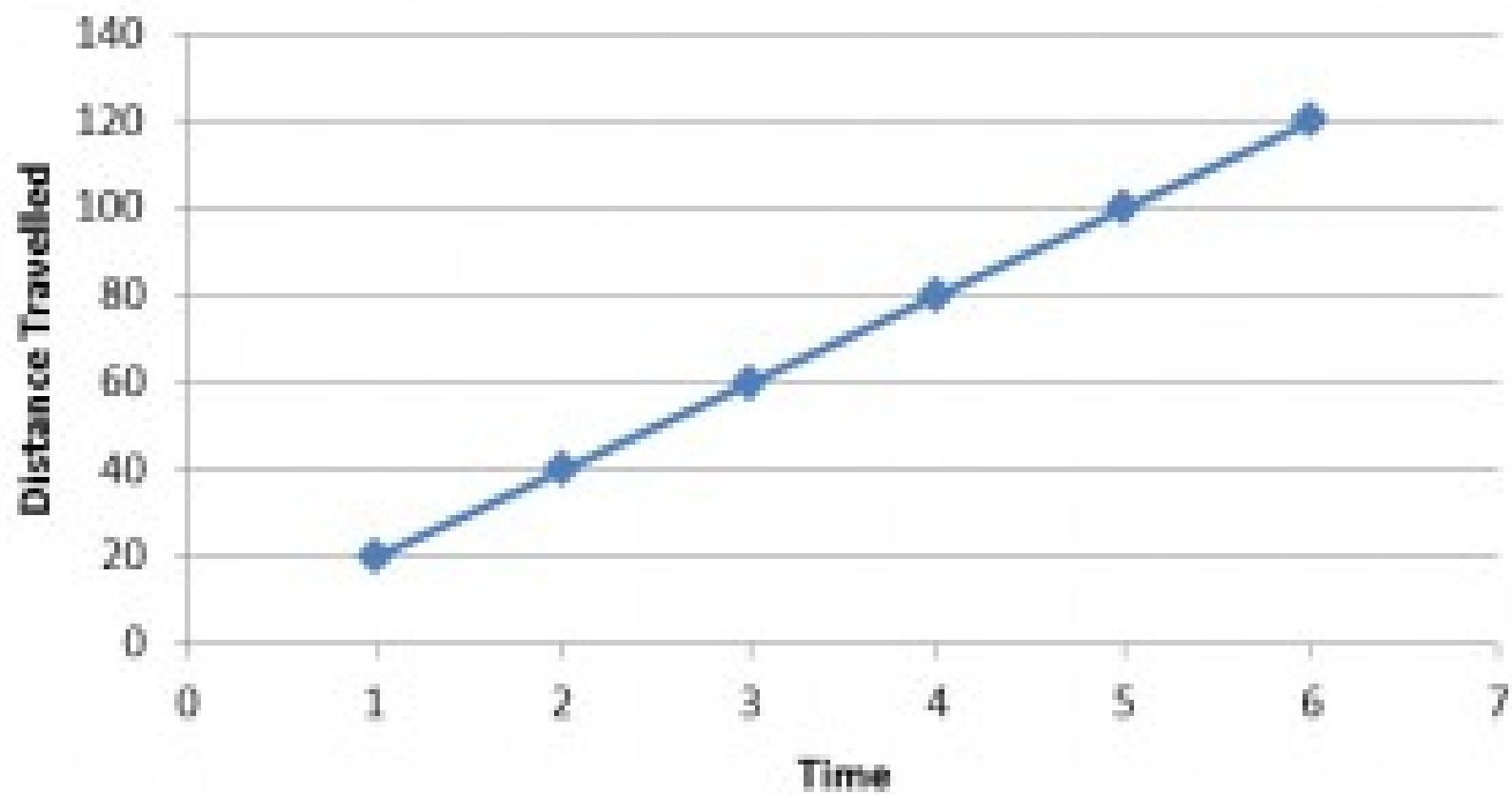


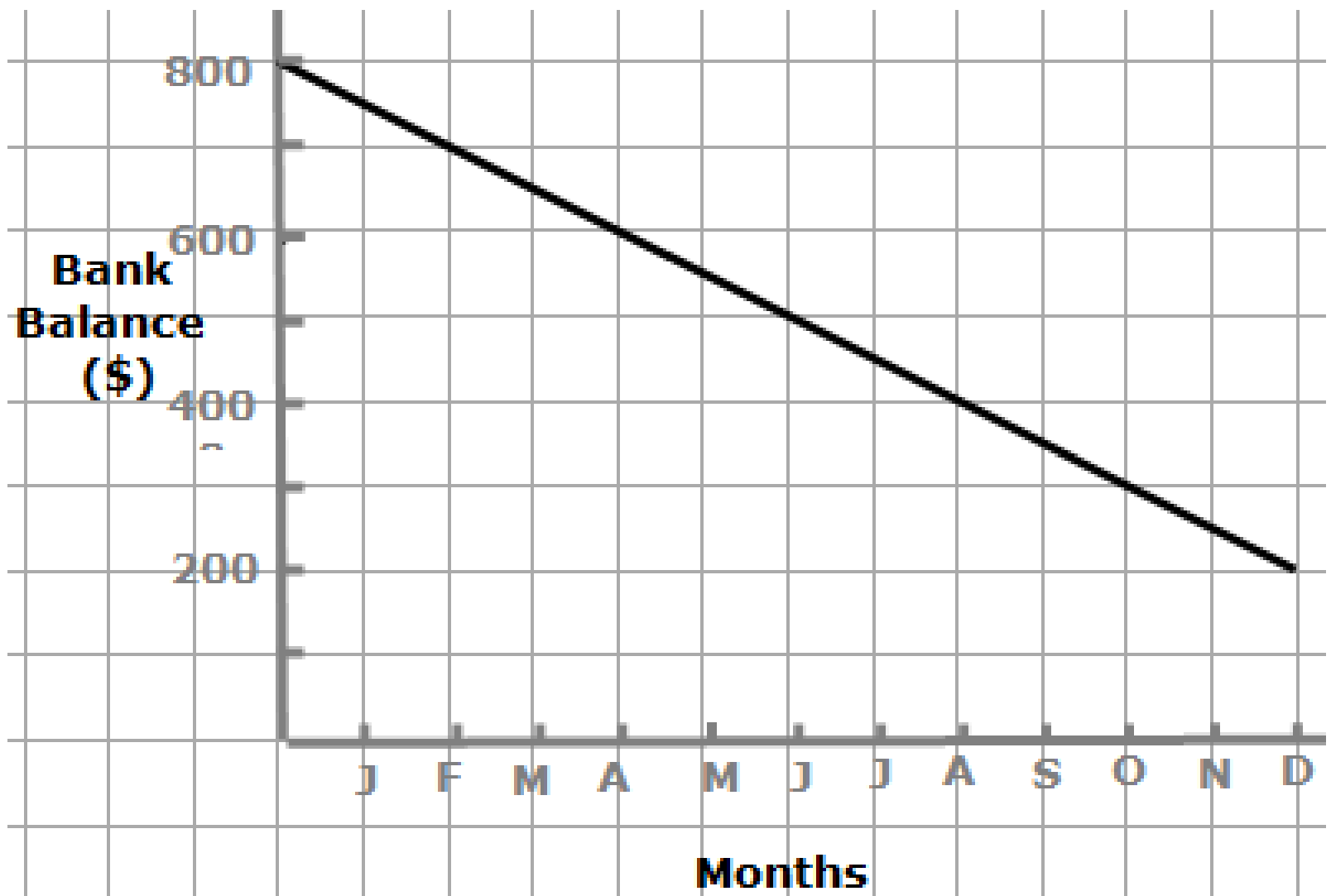
Bacterial Growth at 72°F

(room temperature)









Making a Graph

- Make a data table
- Choose a graph type
- Plot your data
- Do not connect the dots
- Add a trendline if appropriate
 - If linear or exponential
- Give your graph a title
- Label your axes with variable and unit
- Scale your axes

Make a data table

- Put your data in two x-y columns and label it

<i>x</i>	<i>y</i>
-2	-8
-1	-6.5
0	-5
1	-3.5
2	-2

MATH CLASS

Depth (km)	Basalt melting temperature (°C)
0	1100
25	1160
50	1250
100	1400
150	1600

SCIENCE CLASS

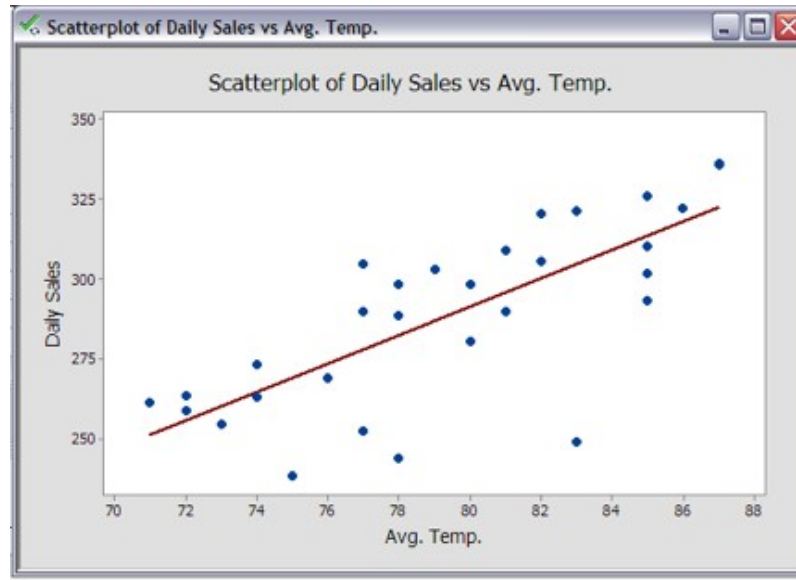
ONLY WRITE UNITS IN TOP BOXES WITH LABELS

Label Your Scatterplots in this Manner

- Give your graph a descriptive title
 - “The relationship between DV (y-axis quantity) and IDV (x-axis quantity)”
- Label your axes with 3 things
 - What it is measuring
 - i.e. temperature, time, mass
 - Units of measurement
 - i.e. Celsius, seconds, grams
 - Show your scale
 - numbers

Add a Trendline

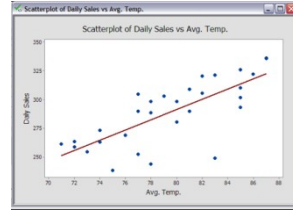
- DO NOT CONNECT THE DOTS
- Add a trendline, linear, exponential or power series as appropriate
- A trendline is the average of your data



Label Your Bar and Pie Charts in this Manner

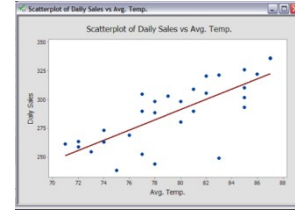
- Give your chart a descriptive title
 - DO NOT call it a relationship
 - Explain what the chart is telling you
 - “Comparison” might be a good title word
- For Bar/Column Charts
 - Label your axes appropriately
 - if necessary provide a key to identify what bars are telling you

Let's Try it In Google Sheets!



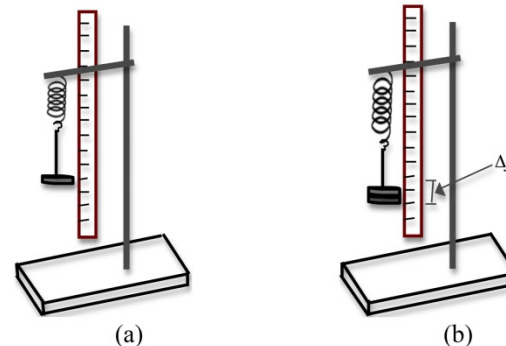
- Everybody open a blank Google Sheets file and follow along with your instructor.

Let's Plot a Relationship!



- The **Force (F)** a spring exerts is directly proportional to the **distance it is stretched** and can be written in the form of a linear $[y = f(x)]$ equation as $F = kx$, where k is the spring constant. Let's make a scatterplot of **$F = kx$** for **$x = 0, .025, .045, .080 \text{ \& } .095 \text{ m}$** where k equals 100.

- $F = 100x$**

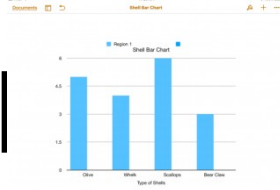


Let's Make a Pie Chart!



- What percentage of students in this class have birthdays during the following quarters?
- Q1 Jan-Mar- 25%
- Q2 Apr-Jun-10%
- Q3 Jul-Sep- 5%
- Q4 Oct-Dec-60%

Let's Make a Bar/Column Graph



- Compare the different cars at the car rental agency when the following cars are available for rent:
- 10 Jeeps, 25 SUVs, 15 Economy Cars, 25 mid-size cars, 5 luxury cars, and 15 convertibles.

Graphing Homework

- Graphing Lab
 - Complete on Google Classroom
 - Due next class
 - DoO not forget to “Turn In”